

Mavolutide Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-876/B

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations,

complaints, and stability data to confirm the continued state of control.

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Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

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Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

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Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

3.2.P.4 Control of Excipients

For the commercial formulation F-876/B, Methocel E5 Premium LV (Hypromellose, release-controlling polymer) is used. Methocel K100M Premium CR was included in the excipient compatibility screen but was not carried forward into the commercial formulation. References to Methocel K4M Premium CR elsewhere in this dossier pertain to development history only.

The approach described herein is consistent with IPEC Excipient Qualification Guide and Ph. Eur. general monograph 2034.

Selection Justification

Methocel E5 Premium LV was selected as the commercial grade of hypromellose on the basis of superior compactibility at low compression forces.

Current Commercial Specification

Component	Function	Grade per current specification
Hypromellose	release-controlling polymer	Methocel E5 Premium LV
Reference		SPEC-EX-6375 Rev 07

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Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission.